# Linux Filesystem

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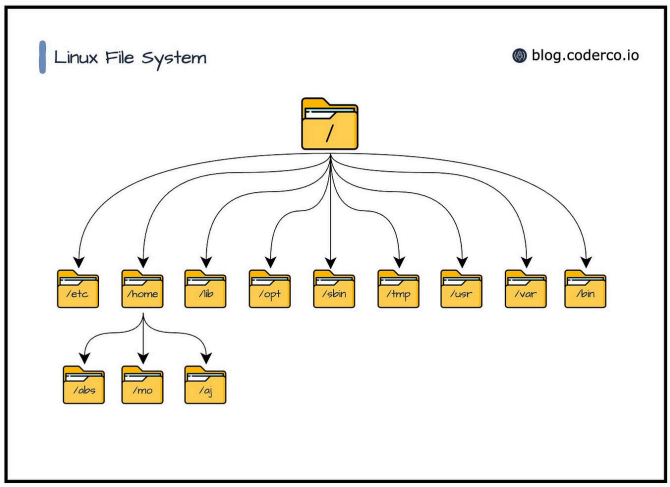
## Introduction to Linux Filesystems

The Linux File System is a structured method of storing and organizing data on a Linux machine. It arranges files in a hierarchical directory format starting from the root directory /. Everything in Linux, including hardware devices, documents and folders, is treated as a file or directory.




## Structure of Linux Filesystems:

The Linux file system architecture is organized into three important layers, each responsible for different functionalities. These layers work together to provide smooth file access, storage, and management.

### 1. Logical File System (LFS)

When we open a file Linux sends the request to the LFS which then understands the file names and then checks if we are authorised to open the file and then respectively controls access to the file like opening, reading and writing. Therefore:-

- Acts as the interface between user applications and the file system.

- Handles key operations like open, read, write, and close.
- Provides security checks, such as permissions and file access control.
- Ensures applications interact with files in a simple and consistent way, without worrying about storage details.

**2. Virtual File System (VFS)**

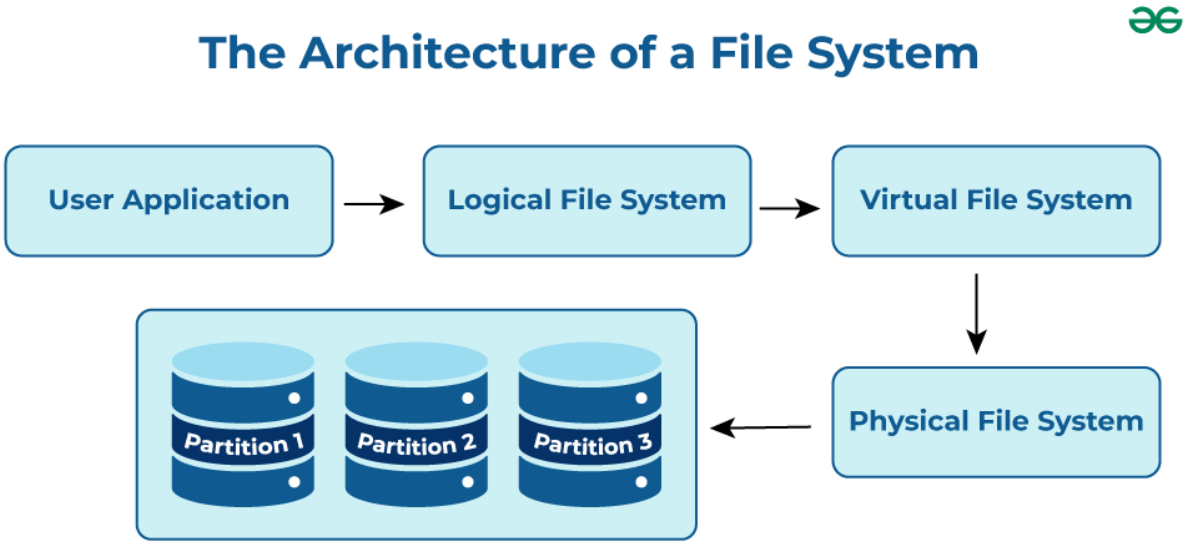
The request is sent from the LFS to the VFS. Different disks on the system support files of different types as mentioned below. VFS hides these differences to make sure that Linux treats all these different file types in the same way (which is why it is also called the translator). So it is the job of the VFS to locate which file system our file is located in.

- Provides a common interface for different file systems (ext4, XFS, FAT32, NTFS, etc.).
- Allows Linux to use multiple file system types at the same time.
- Acts as an abstraction layer, hiding the internal complexities of each file system.
- Makes file operations uniform and compatible, regardless of the underlying file system format.

**3. Physical File System (PFS)**

Finally the request reached the PFS, which interacts with the disk and finds the file and then sends it back.

- Directly interacts with the hardware and disk storage.
- Manages data blocks, inodes, and physical memory allocation.
- Responsible for writing data to the disk and retrieving it efficiently.
- Ensures reliable storage, error handling, and low-level data management.




## Key features of Linux Filesystem

- **Hierarchical Structure:-** Files are organized in a tree-like structure starting from the root directory (/), with folders and subfolders.
- **Everything is a file:-** Almost every component, including hardware devices (hard drives, mouse), is represented as a file, simplifying access and management.
- **Multi-user Support:-** Linux allows multiple users to access the system simultaneously, each with their own files and permissions.
- **Supports multiple file systems:-** like ext4, XFS, FAT32, and NTFS
- **Supports journaling:-** helps prevent data loss and improves recovery after crashes. It logs changes before committing.
- **Strong recovery and permission:-** controls who can access files and what they are allowed to do to the file.

## Key Directories in Linux

- / (Root Directory)

  The top-level directory. All other directories and files start from here.

- /home

  Contains personal folders for each user where their files and documents are stored.

- /bin

  Stores essential command programs used by the system and users.

- /etc

  Contains system configuration files and settings.

- /usr

  Holds user programs, libraries, and application files.

- /var

  Stores variable data such as logs, temporary files, and system data that change frequently.

- /tmp

  Contains temporary files created by programs.

- /dev

  Includes device files that represent hardware devices like disks and keyboards.

- /boot

  Contains files needed to start (boot) the operating system.

## Basic Linux Filesystem commands

- ls (List)

  Displays the files and directories in the current folder.

  ```
  ls [options] [directory]
  ```

- cd (Change Directory)

  Used to move from one directory to another.

  Example: cd /home

  ```
  cd [directory_name]
  ```

- pwd (Print Working Directory)

  Shows the current directory path you are working in.

  ```
  pwd
  ```

- mkdir (Make Directory)

  Creates a new directory (folder).

  Example: mkdir project

  ```
  mkdir [directory_name]
  ```

- rmdir (Remove Directory)

  Deletes an empty directory.

```
rmdir [directory_name]
```

- touch

  Creates a new empty file.

  Example: touch file.txt

```
touch [file_name]
```

- cp (Copy)

  Copies files or directories from one location to another.

```
cp [source_file] [destination]
```

- mv (Move)

  Moves a file to another directory or renames a file.

```
mv [source] [destination]
```

- rm (Remove)

  Deletes files or directories.

```
rm [file_name]
```

- cat (Concatenate)

  Displays the contents of a file on the terminal.

```
cat [file_name]
```

- clear

  Clears the terminal screen.

```
clear
```

- man (Manual)

  Displays the manual/help page for a command.

  Example: man ls

```
man [command_name]
```